

Matrix functions and stability

Complete Crouzeix conjecture

Difficulty: extreme **Importance:** broadly interesting

Status: [Solution claimed](#)

Topic: Matrix-valued polynomial functional calculus

Rating rationale: Historical ratings retained. The uniform bound must survive arbitrary matrix amplification, beyond the scalar functional calculus; it would give sharp bounds for block matrix functions and connect numerical-range estimates to dilation and similarity theory.

Complete resolution claim, 2026-10-06

OpenAI's [mathematics catalog, family 325](#) reports a proof of the sharp constant-two inequality for finite matrix-valued polynomials, and a stronger version for bounded operators on arbitrary complex Hilbert spaces. Its [direct proof manuscript](#) explicitly claims the finite-dimensional bound for all base and coefficient sizes, including point and line-segment numerical ranges. The [Hilbert-space manuscript](#) claims the stronger operator result. Either claimed polynomial inequality would settle the original MF-23 target below; the problem ID, canonical path and statement are retained.

At immutable OpenAI revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, the project publishes a [Lean formalization scope record](#). Its [finite-matrix proof source](#) declares `OAI.DirectCrouzeix.complete_crouzeix : UniversalBound 2`. The [Comparator challenge](#) defines `UniversalBound` with arbitrary positive matrix sizes, degree and complex coefficients, using the Euclidean operator norm and the supremum over the numerical range. The [formalization catalog](#) maps the declaration to this challenge, and its [Comparator configuration](#) permits only Lean's standard foundational axioms.

Evidence level: complete primary-manuscript claim with public Lean proof source and a matching formal statement. This catalog has reviewed the displayed theorem statement and scope, but has not rerun the Lean development or located a dated successful verification log and transitive axiom report for this exact revision. Under this repository's [Lean verification criteria](#), the entry is therefore **Solution claimed**, pending those checks. No external peer review is asserted.

Context and notation

All matrices are complex, and all norms are Euclidean operator norms. For $A \in \mathbb{C}^{n \times n}$ define

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, x^*x = 1\}.$$

If $F(z) = [f_{ij}(z)]_{i,j=1}^m$ has polynomial entries, let $F(A) = [f_{ij}(A)]_{i,j=1}^m$, an $mn \times mn$ block matrix.

Problem statement

Prove or disprove that, for every $n, m \geq 1$, every $A \in \mathbb{C}^{n \times n}$ and every matrix polynomial $F \in \mathbb{C}^{m \times m}[z]$,

$$\|F(A)\|_2 \leq 2 \max_{z \in W(A)} \|F(z)\|_2.$$

This is the complete, rather than scalar, numerical-range spectral-set question. It controls simultaneous polynomial functions and hence block approximation errors with one constant independent of matrix and block sizes.

References

- Michel Crouzeix, *Numerical range and functional calculus in Hilbert space*, *Journal of Functional Analysis* **244**(2), 668–690 (2007). Original source for the complete conjecture, as explicitly attributed in the next source.
- Per Åhag, Rafał Czyż and Jani Virtanen, *Square Functions, Complete Crouzeix Conjecture in Dimension Three, and the Clouâtre–Ostermann–Ransford conjecture*, *arXiv:2608.27346v3*, **9 September 2026**, Introduction, equation (1.2) and **Theorem 1.1**. This current version states the exact target and explicitly distinguishes it from the scalar proofs. It claims the complete bound for $n \leq 3$. Its counterexample reduction leaves only 2×2 matrix polynomials to check when $n = 4$.
- Michel Crouzeix and César Palencia, *The numerical range is a $(1 + \sqrt{2})$ -spectral set*, *SIAM Journal on Matrix Analysis and Applications* **38**(2), 649–655 (2017). The complete universal upper bound is $1 + \sqrt{2}$.

Earlier scope and status check (11 September 2026)

The target is source-stated, not an editorial extension of the scalar conjecture. The August 2026 scalar proof claims by Jin and by Lorist–Schwenninger do not settle it: the September 9 primary manuscript explicitly addresses this distinction. The $n = 3$ result was then a recent preprint claim; the full displayed target remained unresolved even if that claim was accepted. Searches through 11 September 2026 for complete Crouzeix proofs, matrix-valued Crouzeix inequalities and later versions found no full solution. The version was inspected as both HTML and PDF; the inequality is on PDF page 2, immediately before Theorem 1.1.